STQA Practical No. 4

Create Defect Report for Any Application or Web Application

1. AIM

To identify, document, and report defects found in a web application using a structured defect-reporting format.

2. OBJECTIVE

To understand the defect-reporting process by preparing detailed defect reports that include defect IDs, titles, descriptions, severity, priority, preconditions, reproduction steps, expected results, actual results, and defect status.

3. DEFECT REPORT DESCRIPTION

A defect report records a problem found in a software application so that developers can understand, reproduce, and fix it. A good report includes clear steps to reproduce the problem, the expected and actual results, and the environment in which it occurred. Severity describes how much a defect affects the application, while priority describes how urgently it should be fixed. Clear defect reports help developers investigate issues quickly, decide the order of fixes, and let testers confirm the fix later through retesting.

Severity and Priority Scale Used in This Practical

Severity	Meaning
Critical	Application crashes, data loss, or a security breach; no workaround.
Major	Main feature fails or gives wrong results; a workaround may exist but is difficult.
Minor	Small functional issue or display problem; the main feature still works.
Trivial	Cosmetic issue such as a spelling mistake or minor alignment problem.

Priority	Meaning
High	Fix as soon as possible; blocks testing or release.
Medium	Fix in the next planned release or sprint.
Low	Fix when time allows.

4.APPLICATION DETAILS

The defects below belong to a Social Media Web Application. All defects are illustrative examples prepared for this practical. They are not verified bugs in a real application.

4. DETAILED DEFECT REPORTS

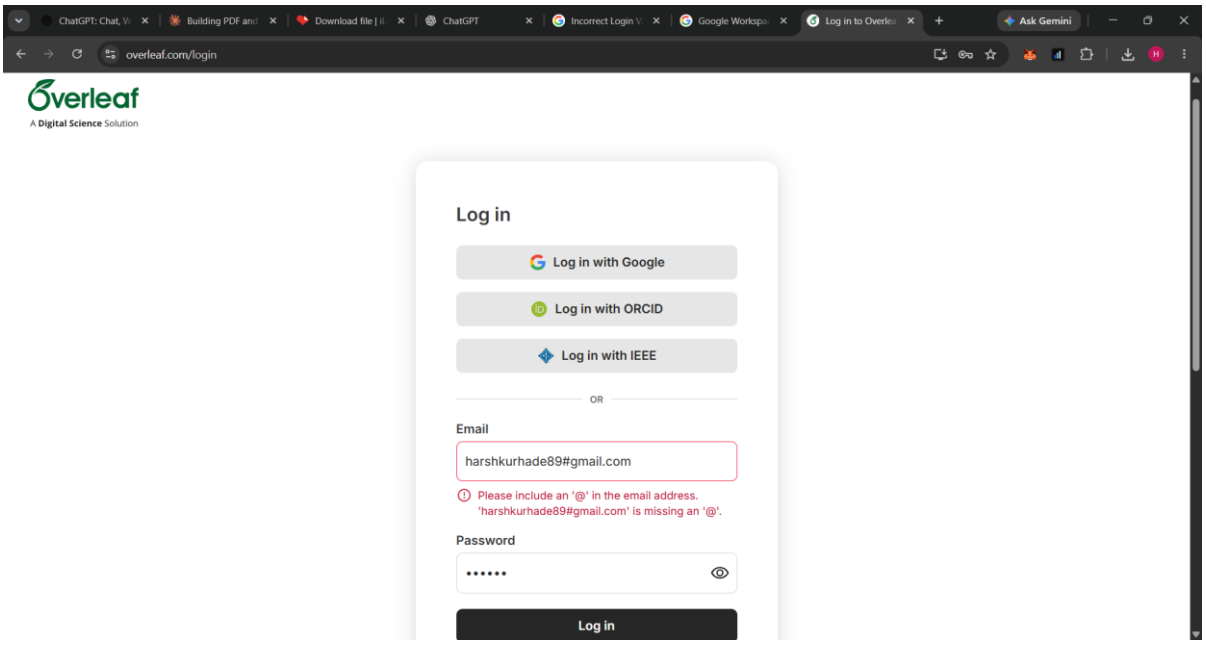
Defect Report 1: Incorrect Login Validation (DEF-001)

Defect ID	DEF-001
Defect Title / Summary	Login Form Does Not Display Proper Validation for Invalid Credentials
Application Name	Social Media Web Application
Module Name	Login / Authentication
Defect Type	Functional Defect
Defect Description	When an invalid username or wrong password is entered on the login page, the application does not show clear validation feedback to the user. The user is not told what went wrong.
Preconditions	A test account or authorized test environment is available.
Test Data	Dummy test credentials only (e.g., username: test_user_01; password: Wrong@9999).
Environment	Web browser on desktop
Severity	Major
Priority	High
Defect Status	Open (illustrative sample)
Reported By	Student (Test Engineer, practical sample)
Date Reported	Not Recorded (sample report)
Assigned To	Not Assigned
Suggested Fix / Developer Comments	Implement proper authentication error handling and show a clear validation message for invalid credentials. Not Assigned. No developer comments recorded.
Retest Status	Not Retested (sample report)

S.No.	**Steps to Reproduce**
1	Open the social media web application.
2	Navigate to the login page.
3	Enter an invalid test email or username.
4	Enter an incorrect test password.
5	Click the Login button.

Expected Result	The application should reject invalid credentials and display an appropriate validation message.
Actual Result	(Illustrative) The application fails to display clear validation feedback.

Figure 1: Sample Login Validation Defect



Defect Report 2: Like Counter (DEF-002)

Defect ID	DEF-002
Defect Title / Summary	Like Counter Does Not Update After Clicking the Like Button
Application Name	Social Media Web Application
Module Name	Social Media Feed / Posts
Defect Type	Functional Defect
Defect Description	When a user clicks the Like button on a post, the button changes state, but the displayed like count does not change.
Preconditions	The user is logged in and a test post is available.
Test Data	Test post by test_user_02 (dummy account).
Environment	Web browser on desktop
Severity	Minor
Priority	Medium
Defect Status	Open (illustrative sample)
Reported By	Student (Test Engineer, practical sample)
Date Reported	Not Recorded (sample report)
Assigned To	Not Assigned
Suggested Fix / Developer Comments	Correct the like-count update logic and make sure the user interface reflects the saved like state. Not Assigned. No developer comments recorded.
Retest Status	Not Retested (sample report)

S.No.	**Steps to Reproduce**
1	Open the social media feed.
2	Locate a test post.
3	Note the current like count.
4	Click the Like button.
5	Observe the button state and the displayed like count.

Expected Result	The Like button should change state and the displayed count should update according to the application's design.
Actual Result	(Illustrative) The button changes state, but the displayed like count does not update.

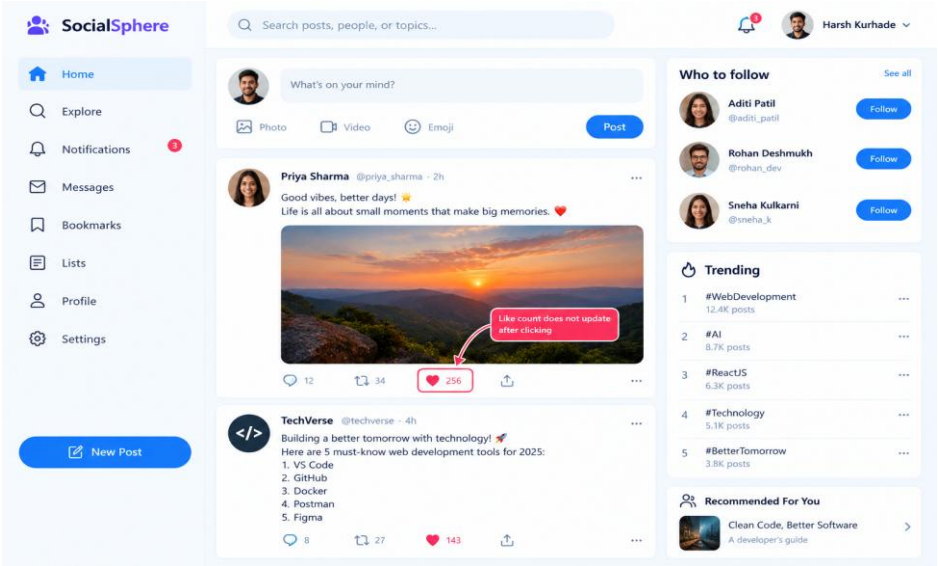


Figure 2: Sample Like Counter Defect (illustrative mock-up)

Defect Report 3: Comment Display (DEF-003)

Defect ID	DEF-003
Defect Title / Summary	Submitted Comment Does Not Appear Under the Post
Application Name	Social Media Web Application
Module Name	Comments
Defect Type	Functional Defect
Defect Description	After a user submits a comment on a post, the submission appears to succeed, but the new comment is not shown in the comments section.
Preconditions	The user is logged in and can comment on a test post.
Test Data	"This is a sample test comment."
Environment	Web browser on desktop
Severity	Major
Priority	High
Defect Status	Open (illustrative sample)
Reported By	Student (Test Engineer, practical sample)
Date Reported	Not Recorded (sample report)
Assigned To	Not Assigned
Suggested Fix / Developer Comments	Investigate the comment-submission API, database persistence, and the user-interface refresh logic. Not Assigned. No developer comments recorded.
Retest Status	Not Retested (sample report)

S.No.	**Steps to Reproduce**
1	Open the social media feed.
2	Select a test post.
3	Enter the sample comment.
4	Click the Submit or Comment button.
5	Check the comments section.

Expected Result	The comment should be saved successfully and displayed under the correct post.
Actual Result	(Illustrative) The comment submission appears successful, but the new comment is not displayed.

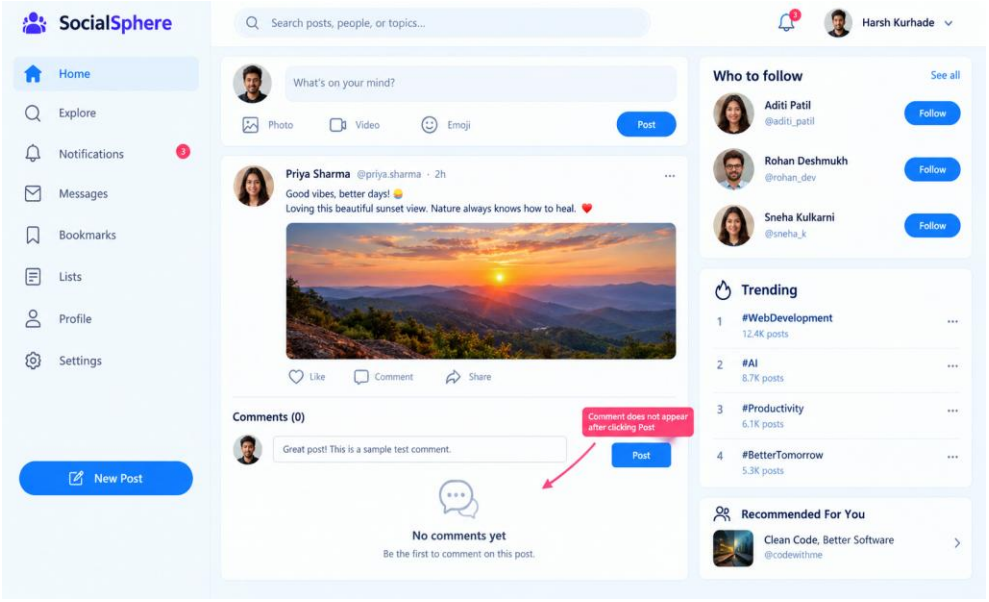


Figure 3: Sample Comment Display Defect (illustrative mock-up)

5. DEFECT SUMMARY TABLE

Defect ID	Defect Title	Module	Severity	Priority	Status
DEF-001	Login Form Does Not Display Proper Validation for Invalid Credentials	Login / Authentication	Major	High	Open (illustrative)
DEF-002	Like Counter Does Not Update After Clicking the Like Button	Social Media Feed / Posts	Minor	Medium	Open (illustrative)
DEF-003	Submitted Comment Does Not Appear Under the Post	Comments	Major	High	Open (illustrative)

6.RESULT

Three sample defect reports have been prepared for the Social Media Web Application. They cover login validation (DEF-001), like-counter update (DEF-002), and comment display (DEF-003). These reports are illustrative. They have not been confirmed by actual testing, so the actual results and statuses should be replaced with real findings if the defects are reproduced.

7.CONCLUSION

Defect reports help testers communicate problems clearly to developers. A well-written report supports reproduction of the issue, classification by severity and priority, and fixing and retesting. Using a consistent template makes it easier to track each defect from the first report until it is closed.